

Detection: Right Number, Wrong Context

Individual component of FaithGuard · M.L Ahamed · Revised plan, 8 October 2026

This document describes one component of FaithGuard, a layer that checks AI answers about financial reports against the evidence they were given, and then sends each answer, fixes it, or withholds it with a reason. This component does the checking: it marks the words in an answer that the evidence does not support, names which financial relation failed, recomputes every number with exact rules, and outputs a calibrated risk.

Nothing in this document has been built or measured yet. Every result below is a hypothesis with a stated way to reject it. All example numbers are illustrative, not results.

1 Purpose

This component tests whether free, automatically mined XBRL errors teach a detector to catch “**right number, wrong context**” mistakes, and whether it still works when moved from US to Sri Lankan reports.

Working title: Right Number, Wrong Context: Relation-Aware Hallucination Detection for Financial Report QA with XBRL-Mined Supervision.

For every answer, the component produces:

1. the **spans** of the answer that the evidence does not support;
2. the **failing relation slot** for each span: entity or scope, metric, period, unit, scale or currency, sign, basis, or missing operand;
3. a **rule verdict** for every numeric claim, recomputed with exact arithmetic and linked to its table cell;
4. a **calibrated risk** and a threshold chosen with Learn-then-Test.

Example

The question is “What was the Group’s profit after tax in FY2025?” The AI answers: “Group profit after tax rose 7.0% to Rs. 12,450 million.”

Detection flags “Rs. 12,450 million” with two slots. *Entity or scope*: it is the Bank’s figure, not the Group’s. *Scale*: the column is in Rs. '000. It also flags “7.0%” as depending on that number. The decision policy then uses these signals to choose an action.

2 Position in the system

Content	
Receives	The question, the frozen evidence (hand-corrected tables in the main test) and the stored AI answer.
Produces	Flagged spans with relation slots; Channel B verdicts and cell links for every numeric claim; a fused, calibrated risk; the accept threshold. These feed the decision policy.
Does not do	It does not choose the action (the decision policy does), does not fix text (the repair component does), and does not retrieve documents (retrieval is frozen).

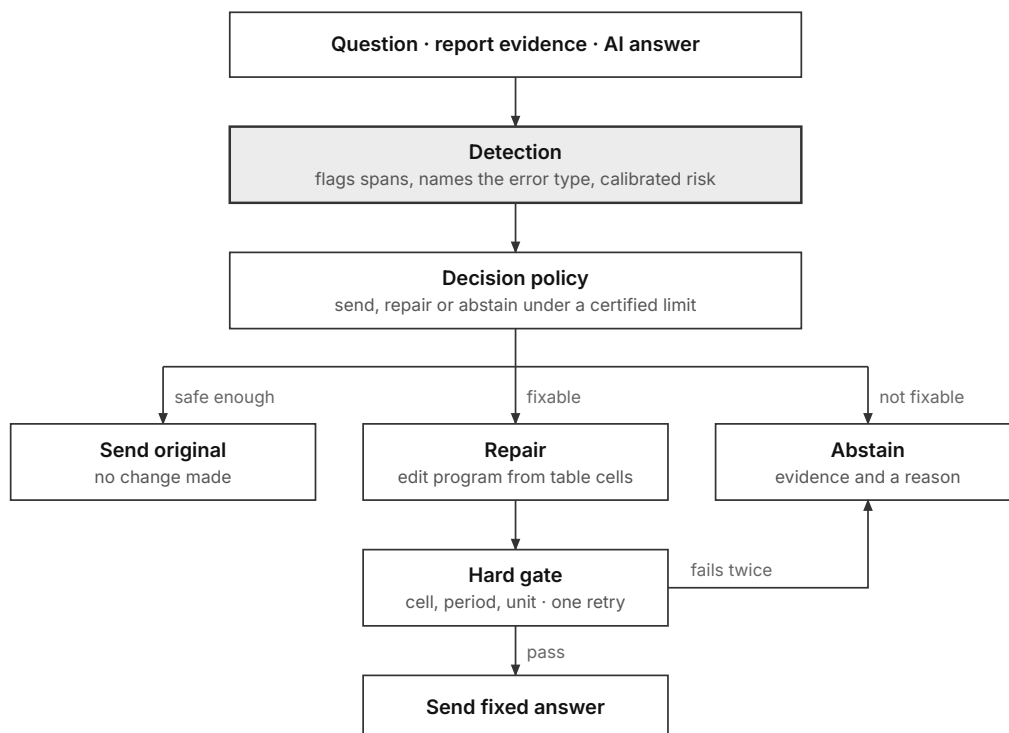


Figure 1. Where the component sits in FaithGuard. The shaded box is this component.

3 Research questions

1. Do XBRL-mined wrong-context negatives raise detection of wrong-context errors in natural answers, at an equal false-alarm rate on clean answers?
2. Where does a learned detector add value over exact rules: Sri Lankan reports without XBRL, claims with no printed number, or correct claims that rules reject?
3. Does a threshold calibrated on US filings hold on Sri Lankan reports, and how many Sri Lankan labels does recalibration need?

4 Component architecture

Two channels check every answer. Their outputs are fused, calibrated and thresholded. XBRL mining and the shift study run offline.

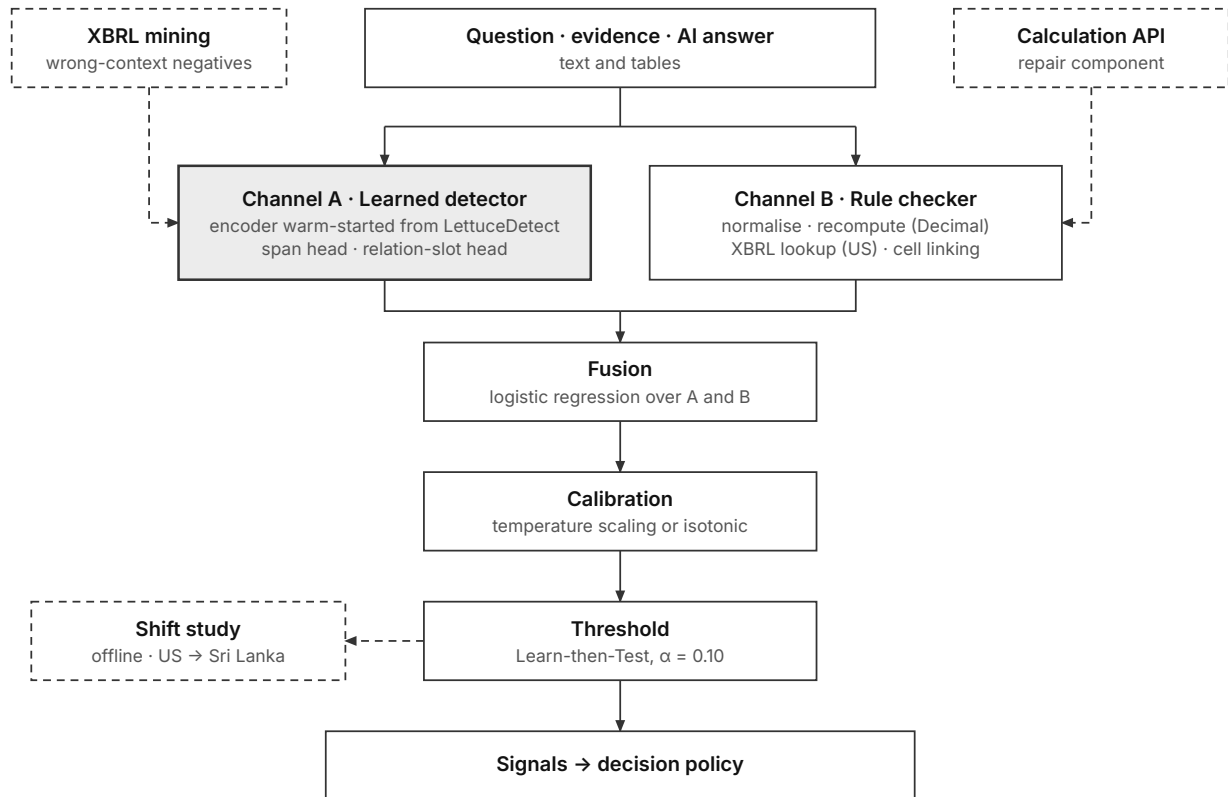


Figure 2. Internal architecture. Channel A (shaded) is the model this component trains. Dashed boxes run offline or belong to another component.

Channel A · Learned detector

One encoder, warm-started from a released LettuceDetect checkpoint (v1-large or the v2 mmBERT encoder, chosen in the week-1 pilot). It has two heads:

- a **span head** that tags unsupported answer tokens;
- a **relation-slot head** that names the failing part: entity or scope, metric, period, unit, scale or currency, sign, basis, or missing operand.

Channel B · Rule checker

It extracts each numeric claim, normalises it and recomputes it with Decimal arithmetic through the shared calculation API. For US filings it looks up XBRL facts. Channel B also links each claim to its table cell by value matching, which replaces a learned evidence pointer.

Fusion and calibration

A logistic regression combines A and B. Temperature scaling or isotonic regression calibrates the result, and Learn-then-Test picks the threshold.

XBRL mining recipe

For each cited fact, find facts with the same value but a different period, segment, entity or concept. Each one becomes a wrong-context negative labelled with the slot that differs. Scale and unit variants (thousands vs millions, percent vs basis points) are added the same way.

5 Data and shared foundation

All three components use one benchmark, built once and frozen, so their results can be compared and combined.

Source	Role	Labels	Licence
RAGTruth	General span training for detection	Human spans	MIT
FinQA	Gold programs for repair targets; injected errors	Gold answers and programs	MIT
TAT-QA	Table-and-text arithmetic targets	Gold answers	CC BY 4.0
US XBRL facts (SEC EDGAR)	Automatic wrong-context negatives; operand checks	Automatic	Public filings
Team benchmark, US and Sri Lanka	Main natural-error test, calibration set, policy outcomes	Human, blinded	Team release terms

FinRank (fresh 2024–25 US questions, CC BY-NC 4.0) is the one optional external test.

Answers and evidence

- **Two generators:** Qwen3.5-9B and Gemma 4 12B-it, both Apache-2.0, run once at 4-bit with llama.cpp on a free T4 and stored. Two vendors are enough for a leave-one-generator-out test.
- **Questions:** 150 US and 250 Sri Lankan question groups from about 12 held-out issuers per country. Each question is written with its gold answer, gold table cells and question type, so code can check numbers later.
- **Evidence:** the main test uses hand-corrected tables, so table-reading errors cannot be mistaken for model errors. One automatic-extraction run, using the best of three tools on 20 tables, is reported as the real-world condition.
- **Retrieval is frozen.** Every method sees the same evidence for the same question.

Rules that keep results honest

- Training, development, calibration and test splits are issuer-disjoint: one company never appears in two splits.
- The test set is locked before any tuning. Hypotheses, the risk target ($\alpha = 0.10$), splits and primary comparisons are pre-registered before the test set is opened.
- One versioned record format is shared by all components. Gold labels live in a separate store that no component reads at decision time.
- Intervals use a bootstrap clustered by issuer. Counts of issuers, questions and answers are always reported separately.

- LLM judges are used for silver training labels only, never for test labels.

Shared tasks owned by this member

- US data: EDGAR download and XBRL parsing with Arelle.
- Label Studio setup and the annotation guide.
- With the other two members: Sri Lankan report collection and table correction (split by sector), and writing about 133 question groups with gold answers.

6 Experiments

Comparison	Question it answers
Channel A with vs without XBRL negatives	RQ1, the main ablation
A alone, B alone, A + B	RQ2: what rules already solve
With vs without the slot head	Does naming the error improve finding it?
US threshold on Sri Lankan data as-is, then recalibrated with 25, 50, 100 and 200 local labels	RQ3: a label-efficiency curve
LettuceDetect v1-large, LettuceDetect v2 encoder, HHEM-2.1-Open, Granite Guardian 4.1-8B	Is the new detector better than released ones?

Metrics: span F1 by slot type, wrong-context recall, false-alarm rate on clean answers (a headline metric), AUROC, Brier score and ECE, and coverage at certified selective risk ($\alpha = 0.10$).

7 Hypotheses

Hypothesis	Confirmed if	Rejected if
H1. XBRL negatives raise wrong-context recall	Higher recall at an equal clean false-alarm rate	False alarms rise or recall does not move
H2. A + B beats B alone on Sri Lankan reports	Higher span F1, intervals not overlapping	No gain; rules suffice, reported as such
H3. The US threshold fails on Sri Lanka; about 100 local labels restore it	Risk above α before, within α after	Holds without recalibration; also reported

8 Scope and compute

- **Minimum publishable result:** the trained detector, the XBRL ablation, the clean false-alarm rate and the recalibration curve, all on human-labelled natural errors. This counts even if no baseline is beaten.
- **Stretch:** a consistency loss against simple augmentation; the observer-probe signal (Channel C).
- **Stands alone:** it needs only the shared benchmark, not repair or the policy.

- **Compute:** a 0.3–0.4B encoder at 1–2K tokens in fp16 with gradient checkpointing fits a free Kaggle T4 or P100. No university GPU is needed.

Work on this member's quota	Rough GPU hours
Baseline detectors, inference only (Granite Guardian 8B at 4-bit)	3–6
Detector training: main model and ablations	25–45
Share of pilots, debugging and reruns (15–25 hours across the team)	—

These figures are rough guesses, not measurements. The week-1 pilots replace them with real timings. Runs are kept to a few hours, resumable, with a checkpoint at least every hour.

9 Relation to existing work

Defensible claim: XBRL facts are used as a *training* signal for a span detector. VeriFin uses XBRL only at inference time. The detector also names the failing financial relation, and it is tested across a real US-to-Sri-Lanka shift with a measured recalibration cost.

Claims to avoid: the first financial hallucination detector; the first typed or span-plus-evidence detector.

Realistic first target: FinNLP 2027 or SLIIT ICAC 2027.

10 Risks and cut list

Risk	Early warning	Fallback
XBRL negatives do not help	No recall gain on the dev set	Report as a negative result; the slot head and shift study still stand
Labelling is slower than 4 min per item	Pilot rate above 6 min	Cut US groups to 120 first; keep all Sri Lankan groups
Generators make very few errors	Pilot error rate below about 10%	Report it as a finding that bounds the benefit; add harder question types
A teammate's part is late	A checkpoint is missed	Nothing to switch: this component needs only the shared benchmark
A new paper overlaps the contribution	Monthly literature refresh	Narrow the claim and cite it; never hide it
Free GPU quota is cut or runs out	Hours used pass the budget	Use Sharaf's spare quota; switch between Kaggle and Colab; apply the cut list

Cut list items for this component, in team order: Granite Guardian baseline (2nd), generator-shift test (6th), slot-head ablation (7th). **Never cut:** the human-labelled natural test set, the clean false-alarm rate and the main XBRL ablation.

11 Plan

By	Milestone
18 Oct 2026	Week-1 pilot done: LettuceDetect checkpoint chosen; GPU quota recorded
31 Oct 2026	Thin end-to-end demo on 20–40 items, rules only
19 Oct – 30 Nov 2026	XBRL mining and the Channel B rule checker
30 Nov 2026	Pilot labels done and the real labelling rate known
1 Dec 2026 – 31 Jan 2027	Joint training and ablations
31 Jan 2027	Test set locked; pre-registration filed
15 Feb 2027	Detector model frozen
1 Feb – 31 Mar 2027	US-to-Sri-Lanka shift study
31 May 2027	Paper draft
30 Jun 2027	Final report and demo

References

1. C. Niu et al. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. arXiv:2401.00396, 2024.
2. Á. Kovács and G. Recski. LettuceDetect: A Hallucination Detection Framework for RAG Applications. arXiv:2502.17125, 2025.
3. A. Angelopoulos et al. Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. arXiv:2110.01052, 2021.
4. Vectara. HHEM-2.1-Open hallucination evaluation model. Hugging Face model card, [vectara/hallucination_evaluation_model](#).

VeriFin and Granite Guardian 4.1-8B are named in the text; their full citations are added once each source has been opened and logged in the team’s related-work notes.